SQL Data Analysis Report

Bird Species Observation Dataset

Introduction

This report presents an SQL-based analysis of the `Bird_Observation_DataSet`, which contains detailed records of bird species observed across multiple habitats, primarily forests and grasslands. The dataset includes information on species identification, observation time, location characteristics, environmental conditions, and observer details. The main objective of this analysis is to understand patterns in bird species distribution, diversity, behavior, and habitat preferences, along with the influence of environmental factors on bird activity.

The analysis is conducted using the `bird_data` table, which includes key attributes such as `Common_Name`, `Scientific_Name`, `Location_Type`, `Plot_Name`, `Date`, `Start_Time`, `Observer`, `ID_Method`, `Flyover_Observed`, `Sky`, `Wind`, and `Disturbance`. Additional fields such as `PIF_Watchlist_Status` and `Regional_Stewardship_Status` are used to derive conservation-related insights. Derived temporal features like month and hour are also utilized to analyze seasonal trends and daily activity patterns.

The insights generated from this analysis aim to support biodiversity conservation, habitat management, ecological monitoring, and data-driven decision-making for environmental sustainability.

SQL Queries and Insights

Overall Observations and Species Diversity

Objective: Determine the total number of observations, unique bird species, and total survey locations.

```sql
SELECT
   COUNT(*) as Total_Observations,
   COUNT(DISTINCT "Scientific_Name") as Unique_Species,
   COUNT(DISTINCT "Plot_Name") as Total_Locations
FROM bird_data;
```

	total_observations bigint	unique_species bigint	total_locations bigint
1	15372	127	609

Figure 1: total number of observations, unique bird species, and total survey locations

Observation: The query provides a summary of the dataset, including the total number of bird observations, the number of distinct species recorded, and the number of unique locations (plots) surveyed.

Insight: A higher number of unique species indicates greater biodiversity, while a larger number of locations reflects wider spatial coverage of the study.

Implication: These metrics serve as foundational indicators for ecological analysis and help assess the richness and coverage of the dataset before performing deeper analysis.

Top 10 Most Frequently Observed Bird Species

Objective: Identify the most commonly observed bird species based on observation count.

```
SELECT "Common_Name", COUNT(*) as Observation_Count
FROM bird_data
GROUP BY "Common_Name"
ORDER BY Observation_Count DESC
LIMIT 10;
```

	Common_Name text	observation_count bigint
1	Northern Cardinal	1125
2	Carolina Wren	993
3	Red-eyed Vireo	738
4	Eastern Tufted Titmou...	720
5	Indigo Bunting	611
6	Eastern Wood-Pewee	574
7	Field Sparrow	492
8	Red-bellied Woodpeck...	489
9	American Robin	470
10	Acadian Flycatcher	462

Figure 2: Top 10 Most Frequently Observed Bird Species

Observation: The query returns the top 10 bird species with the highest number of observations in the dataset.

Insight: These species are likely the most abundant or easily detectable in the surveyed regions, indicating their dominance in the ecosystem.

Implication: Understanding frequently observed species helps in tracking population trends, identifying dominant species, and supporting biodiversity monitoring efforts.

Temporal Analysis: Observations by Month

Objective: Analyze the distribution of bird observations across different months to identify seasonal trends.

```
SELECT EXTRACT(MONTH FROM CAST("Date" AS DATE)) AS "Month_Number",
   COUNT(*) AS "Observations"
FROM bird_data
WHERE "Date" IS NOT NULL
GROUP BY EXTRACT(MONTH FROM CAST("Date" AS DATE))
ORDER BY "Month_Number";
```

	Month_Number numeric	Observations bigint
1	5	4864
2	6	6211
3	7	4297

Figure 3: Observations by Month

Observation: The query groups observations by month, showing how bird sightings vary throughout the year.

Insight: Variations in monthly observations indicate seasonal patterns, such as migration periods, breeding seasons, or changes in bird activity levels.

Implication: Understanding seasonal trends helps in planning field surveys, conservation strategies, and monitoring bird populations during critical periods of the year.

Temporal Analysis: Observations by Hour of Day

Objective: Identify peak hours of bird activity based on observation counts.

```
SELECT SUBSTRING(CAST("Start_Time" AS TEXT) FROM 1 FOR 2) AS "
   Hour_of_Day",
      COUNT(*) AS "Observations"
FROM bird_data
WHERE "Start_Time" IS NOT NULL
GROUP BY SUBSTRING(CAST("Start_Time" AS TEXT) FROM 1 FOR 2)
ORDER BY "Observations" DESC;
```

Observation: The query extracts the hour from the observation start time and counts the number of observations for each hour of the day.

Insight: Higher observation counts are typically recorded during early morning hours, reflecting peak bird activity periods such as the dawn chorus.

	Hour_of_Day text	Observations bigint
1	07	4014
2	06	3837
3	08	3196
4	09	2016
5	05	1288
6	10	1021

Figure 4: Observations by Hour of Day

Implication: Identifying peak activity hours helps optimize survey timing, ensuring more efficient data collection and better detection of bird species.

Spatial Analysis: Forest vs Grassland Comparison

Objective: Compare bird observations across major habitat types such as forest and grassland.

```
SELECT "Location_Type", COUNT(*) AS "Count"
FROM bird_data
WHERE "Location_Type" IN ('Forest', 'Grassland')
GROUP BY "Location_Type"
ORDER BY "Count" DESC;
```

	Location_Type text	Count bigint
1	Forest	8546
2	Grassland	6826

Figure 5: Forest vs Grassland Comparison

Observation: The query shows the number of bird observations recorded in forest and grassland habitats.

Insight: Differences in observation counts indicate variation in bird activity and species richness between habitat types.

Implication: Understanding habitat-based distribution helps in identifying critical ecosystems and supports targeted conservation and habitat management strategies.

Top 10 Biodiversity Hotspots (Plot-Level Analysis)

Objective: Identify the plots with the highest number of bird observations.

```
SELECT "Plot_Name", COUNT(*) AS "Observation_Count"
FROM bird_data
GROUP BY "Plot_Name"
ORDER BY "Observation_Count" DESC
LIMIT 10;
```

	Plot_Name text	Observation_Count bigint
1	ANTI-0163	54
2	HAFE-0067	52
3	ANTI-0028	52
4	MANA-0048	50
5	ANTI-0016	49
6	MONO-0057	48
7	ANTI-0184	48
8	ANTI-0034	48
9	ANTI-0161	48
10	MONO-0037	48

Figure 6: Top 10 Biodiversity Hotspots

Observation: The query highlights the top 10 plots with the highest observation counts in the dataset.

Insight: Plots with higher observation counts can be considered biodiversity hotspots, indicating areas with greater bird activity or species richness.

Implication: These hotspots are important for conservation planning, habitat protection, and targeted ecological monitoring efforts.

Environmental Analysis: Sky Conditions Distribution

Objective: Analyze how different sky conditions affect bird observations.

```sql
SELECT "Sky" AS "Sky_Condition", COUNT(*) AS "Observations"
FROM bird_data
GROUP BY "Sky"
ORDER BY "Observations" DESC;
```

	Sky_Condition text	Observations bigint
1	Partly Cloudy	6172
2	Clear or Few Clou...	5333
3	Cloudy/Overcast	2917
4	Fog	598
5	Mist/Drizzle	352

Figure 7: Sky Conditions Distribution

Observation: The query provides the distribution of bird observations under different sky conditions.

Insight: Certain sky conditions (such as clear or partly cloudy) tend to have higher observation counts, indicating favorable visibility and bird activity.

Implication: Understanding the influence of weather conditions helps improve survey planning and ensures more accurate and consistent bird observations.

Environmental Analysis: Wind Conditions

Objective: Analyze how wind conditions influence bird observations.

```sql
SELECT "Wind" AS "Wind_Condition", COUNT(*) AS "Observations"
FROM bird_data
GROUP BY "Wind"
ORDER BY "Observations" DESC;
```

Observation: The query shows the distribution of bird observations across different wind conditions.

Insight: Higher observations are typically recorded during calm or low-wind conditions, as birds are more active and easier to detect.

	Wind_Condition text	Observations bigint
1	Light air movement (1-3 mph) smoke dri...	7636
2	Calm (< 1 mph) smoke rises vertically	4210
3	Light breeze (4-7 mph) wind felt on face	3161
4	Gentle breeze (8-12 mph), leaves in mot...	365

Figure 8: Wind Conditions

Implication: Wind conditions should be considered when planning surveys, as strong winds may reduce observation accuracy and bird activity.

Environmental Analysis: Disturbance Impact

Objective: Examine the effect of environmental or human disturbance on bird observations.

```
SELECT "Disturbance" AS "Disturbance_Level", COUNT(*) AS "
   Observations"
FROM bird_data
GROUP BY "Disturbance"
ORDER BY "Observations" DESC;
```

	Disturbance_Level text	Observations bigint
1	No effect on count	7525
2	Slight effect on count	5840
3	Moderate effect on co...	1577
4	Serious effect on count	430

Figure 9: Disturbance Impact

Observation: The query categorizes observations based on different disturbance levels.

Insight: Lower disturbance levels generally correspond to higher bird sightings, indicating that birds prefer stable and undisturbed environments.

Implication: Reducing disturbances in key habitats can support biodiversity conservation and improve ecosystem health.

Behavior Analysis: Identification Method (Auditory vs Visual)

Objective: Compare bird identification methods used during observations.

```
SELECT "ID_Method", COUNT(*) AS "Count"
FROM bird_data
GROUP BY "ID_Method"
ORDER BY "Count" DESC;
```

	ID_Method text	Count bigint
1	Singing	9621
2	Calling	3941
3	Visualization	1808
4	Unknown	2

Figure 10: Identification Method

Observation: The query shows the frequency of different identification methods such as auditory (calls/songs) and visual sightings.

Insight: One method may dominate depending on habitat conditions and species behavior, with auditory detection often being common in dense vegetation.

Implication: Understanding identification methods helps improve survey accuracy and highlights the need for observer training in both techniques.

Behavior Analysis: Flyover vs Ground Activity

Objective: Distinguish between birds observed flying overhead and those actively using the habitat.

```
SELECT "Flyover_Observed" AS "Is_Flyover", COUNT(*) AS "Count"
FROM bird_data
GROUP BY "Flyover_Observed";
```

	Is_Flyover boolean	Count bigint
1	false	14683
2	true	689

Figure 11: Flyover vs Ground Activity

Observation: The query separates observations into flyover and non-flyover categories.

Insight: Grounded observations indicate active habitat usage, while flyovers represent transient movement.

Implication: Habitat-based conservation efforts should prioritize areas with higher grounded bird activity.

Observer Analysis: Top 10 Most Active Observers

Objective: Identify observers who contributed the most data.

```
SELECT "Observer", COUNT(*) AS "Total_Records_Submitted"
FROM bird_data
GROUP BY "Observer"
ORDER BY "Total_Records_Submitted" DESC
LIMIT 10;
```

	Observer text	Total_Records_Submitted bigint
1	Elizabeth Osw...	5763
2	Kimberly Serno	5346
3	Brian Swimelar	4263

Figure 12: Top 10 Most Active Observers

Observation: A small number of observers contribute a large portion of the dataset.

Insight: Data collection may be influenced by observer effort and experience.

Implication: Encouraging broader participation can improve dataset diversity and reduce observer bias.

Conservation Analysis: PIF Watchlist Species

Objective: Identify sightings of species listed in the PIF Watchlist.

```
SELECT "Common_Name", "Scientific_Name", COUNT(*) AS "Sighting_Count
    "
FROM bird_data
WHERE "PIF_Watchlist_Status" = 'True'
GROUP BY "Common_Name", "Scientific_Name"
ORDER BY "Sighting_Count" DESC;
```

	Common_Name text	Scientific_Name text	Sighting_Count bigint
1	Wood Thrush	Hylocichla mustelina	309
2	Worm-eating Warbler	Helmitheros vermivorus	31
3	Prairie Warbler	Setophaga discolor	25
4	Cerulean Warbler	Setophaga cerulea	7
5	Kentucky Warbler	Oporornis formosus	2
6	Willow Flycatcher	Empidonax traillii	2
7	Blue-winged Warbler	Vermivora cyanoptera	1
8	Red-headed Woodpec...	Melanerpes erythrocephal...	1

Figure 13: PIF Watchlist Species

Observation: The query highlights species of conservation concern and their observation frequency.

Insight: Frequent sightings may indicate important habitats or stable populations of at-risk species.

Implication: These findings support targeted conservation planning and monitoring of vulnerable species.

Conservation Analysis: Regional Stewardship Status

Objective: Analyze species based on their regional conservation priority.

```
SELECT "Regional_Stewardship_Status" AS "Stewardship_Status", COUNT
    (*) AS "Count"
FROM bird_data
GROUP BY "Regional_Stewardship_Status"
ORDER BY "Count" DESC;
```

	Stewardship_Status boolean	Count bigint
1	false	11482
2	true	3890

Figure 14: Regional Stewardship Status

Observation: The dataset includes species with varying levels of regional stewardship importance.

Insight: Certain species require more focused conservation efforts due to their ecological significance.

Implication: This analysis helps prioritize conservation resources and supports biodiversity management strategies.

Conclusion

The SQL-based analysis of the `Bird_Observation_DataSet` provides valuable insights into bird species distribution, behavior, and environmental interactions across forest and grassland ecosystems. The study highlights key patterns in species diversity, temporal trends, habitat preferences, and the influence of environmental factors such as sky conditions, wind, and disturbance on bird observations.

Temporal analysis revealed clear seasonal and daily activity patterns, with higher observations during specific months and peak activity during early morning hours. Spatial analysis identified biodiversity hotspots and emphasized differences in bird activity between habitat types, reinforcing the ecological importance of both forests and grasslands.

Behavioral analysis demonstrated the significance of identification methods and distinguished between flyover and habitat-dependent species activity. Observer-based insights also indicated potential biases in data collection, emphasizing the need for broader participation.

Importantly, the conservation-focused analysis identified species listed under the PIF Watchlist and highlighted regional stewardship priorities, providing direction for targeted conservation efforts.

Overall, this analysis supports data-driven decision-making in biodiversity conservation, habitat management, and ecological monitoring. The findings can be further enhanced through visualization tools such as Power BI or Streamlit dashboards, enabling interactive exploration and more effective communication of insights to stakeholders.